

SuryaVayu AI Theme 10

SuryaVayu AI - Theme 10 Solution Document

Problem

KREDL/KSPDCL operators need day-ahead and intraday visibility of renewable output across distributed solar and wind plants, with explicit uncertainty and actionable alerts for ramp or curtailment risk.

Approach

SuryaVayu AI is a Next.js + Prisma control-room application that combines:

1. 90-day hourly synthetic plant generation and weather data.
2. Pure JavaScript forecasting (no Python runtime dependency) with:
 - solar diurnal baseline from past 30 days,
 - weather adjustments (cloud and temperature derating),
 - wind power curve behavior,
 - scenario multipliers (BASE/OPTIMISTIC/PESSIMISTIC),
 - horizon-aware confidence intervals.
3. Forecast quality analytics with MAPE/RMSE/MBE.
4. Alerting for ramp events, curtailment risk, persistent under/over forecast, and offline plants.
5. Model version history with mock retrain flow.

Data Model

Prisma models:

- Plant with enums PlantType and PlantStatus.
- WeatherReading and Generation for time series.
- Forecast with ScenarioType.
- ForecastAccuracy for MAPE/RMSE/MBE.
- Alert with AlertType and AlertSeverity.
- ModelVersion for versioned forecast strategies.

APIs

- GET /api/plants
- GET /api/plants/[id]
- GET /api/plants/[id]/history
- GET /api/plants/[id]/forecast
- GET|POST /api/forecasts/generate
- GET|POST /api/forecasts/accuracy
- GET /api/alerts
- PUT|POST /api/alerts/[id]/ack
- GET /api/dashboard/overview
- GET /api/models/versions
- POST /api/models/retrain

UI Pages

- / Grid control room dashboard with Tremor cards, area chart, map, and alert feed.
- /plants Plant table with type/status controls.
- /plants/[id] Money-shot detail page with tabs: Forecast, History, Weather, Accuracy, Alerts.
- /alerts Grid-wide alert operations page.
- /models Model lifecycle and retrain page.

Seed Design

- scripts/generate-mock-plants.ts: 6 solar + 3 wind plants across Karnataka.
- scripts/generate-mock-timeseries.ts: 90-day hourly generation and weather with diurnal behavior and injected curtailment.
- scripts/seed-demo.ts: loads DB, generates forecasts, computes accuracy, creates alerts, and inserts two model versions.

Mock AI

USE MOCK_AI=true enables deterministic natural-language forecast variance explanation in lib/ai.ts.

Architecture Diagram

- Source: docs/diagrams/architecture.mmd
- Rendered artifact: docs/diagrams/architecture.svg

Known Limitations

- PNG diagram and PDF rendering require local CLI execution (npx @mermaid-js/mermaid-cli, pandoc, headless Chrome).
- Verification gates and git push/repo creation require shell access.

Regenerating the PDF

```
pandoc docs/solution-document.md -o /tmp/suryavayu-solution.html --  
    standalone --metadata title="SuryaVayu AI Theme 10"  
"/Applications/Google Chrome.app/Contents/MacOS/Google Chrome" --  
    headless --disable-gpu --no-pdf-header-footer \  
    --print-to-pdf="$(pwd)/docs/solution-document.pdf"  
    "file:///tmp/suryavayu-solution.html"
```